

TCS3351 Cryptography and Data Security

Tutorial 1

Introduction: Security Goals, Services, Mechanisms, Cryptosystems

1. What do you mean by CIA Triad? Briefly explain each security goal with an example.
2. What is the difference between passive and active security attacks? State the general approach that can be used for handling each type of these attacks.
3. Give an example of the type of attacks that will threaten availability and explain briefly.
4. Briefly explain the meaning of the following terms related to security services:
 - (i) authentication
 - (ii) non-repudiation
5. Briefly discuss the basic model for network security and explain how person A can transmit the information in a secure manner to person B using this model.
6. Briefly describe the model for network access security with typical measures to be taken for security.
7. Identify the type of security attack in the following example and briefly explain the reason for your answer. When purchasing an item from the seller, person A gave an instruction to the bank to deduct RM 500.

Later person A found that instead of RM500, RM5000 was deducted from the account. How can you avoid this type of attack?

8. Briefly explain the differences between symmetric and asymmetric cryptography models.
9. Answer the following questions with respect to symmetric cryptography.

In a cricket team, there are '11' players.

 - (i) How many keys are needed if all the players need to send secret messages to each other?
 - (ii) How many keys are needed if everyone trusts the team's coach? If a player needs to send a message to another player, he/she first sends it to the coach, and the coach then sends the message to the other student.
10. Repeat Question 9, assuming the usage of asymmetric cryptography.

References:

- Behrouz A Forouzan, Debdeep Mukhopadhyay, “Cryptography and Network Security”, Third Edition, Mc-Graw Hill Education, 2015.
- William Stallings and Lawrie Brown, “Computer Security: Principles and Practice”, Fourth Edition, Prentice Hall, 2018.
- William Stallings, “Cryptography and Network Security – Principles and Practice”, Seventh Edition, Prentice Hall, 2017.
- Atul Kahate, "Cryptography and Network Security", Fourth Edition, McGraw-Hill Education, 2019.